

Q1.
 $\Rightarrow$ for $\lambda_{mkt} = \frac{1}{1^T \Sigma^{-1} (\mu - r_f \mathbf{1})}$ — Eq 40

Weights are given by:

$$\omega = \lambda \Sigma^{-1} (\mu - r_f \mathbf{1}), \quad \omega_0 = 1 - \lambda 1^T \Sigma^{-1} (\mu - r_f \mathbf{1})$$

Replacing λ_{mkt}

$$\begin{aligned} \omega &= \frac{\Sigma^{-1} (\mu - r_f \mathbf{1})}{1^T \Sigma^{-1} (\mu - r_f \mathbf{1})} & \omega_0 &= \frac{1 - \frac{1^T \Sigma^{-1} (\mu - r_f \mathbf{1})}{1^T \Sigma^{-1} (\mu - r_f \mathbf{1})}}{1} \\ &= \lambda_{mkt} (\Sigma^{-1} (\mu - r_f \mathbf{1})) & &= 1 - 1 \\ &= \omega_{mkt} \dots \text{from eq (41)} & &= 0 \\ & & & \downarrow \\ & & & \text{Risk-free asset} = 0 \end{aligned}$$

Thus proved!

The efficient frontier & line meet at the market portfolio

problem 1
(32)

(33) → Efficient frontier

$$\mu_p = \frac{a}{c} + \frac{d}{c} \lambda_{mkt} \quad \frac{d\mu_p}{d\lambda} = \frac{d}{c}$$

$$\sigma_p^2 = \frac{1}{c} + \frac{d}{c} \lambda_{mkt}^2 \quad \frac{d\sigma_p}{d\lambda} = \frac{d}{c} \lambda_{mkt}$$

$$\therefore \frac{d\mu_p}{d\sigma_p} = \frac{\sigma_p}{\lambda_{mkt}} \quad \text{--- (A)}$$

(39) → CML

$$\mu_p - r_f = \sigma_p \sqrt{h}$$

$$\Rightarrow \frac{d\mu_p}{d\sigma_p} = \sqrt{h} = \frac{\mu_p - r_f}{\sigma_p} \quad \text{--- (B)}$$

$$\mu_p = r_f + h\lambda \quad \left. \begin{array}{l} \sigma_p^2 = h\lambda^2 \end{array} \right\} \text{--- (37)}$$

$$\sqrt{h} = \frac{\sigma_p}{\lambda_{mkt}} \quad \text{--- (C)} \quad \lambda = \lambda_{mkt}$$

~~scribbles~~

(A) ⇒ slope = $\frac{\sigma_p}{\lambda_{mkt}}$

(B) ⇒ slope = $\sqrt{h}$

(C) ⇒ $\sqrt{h} = \frac{\sigma_p}{\lambda}$

∴ B, C ⇒ slope of CML = $\sqrt{h} = \frac{\sigma_p}{\lambda_{mkt}}$

A ⇒ slope of Efficient frontier = $\frac{\sigma_p}{\lambda_{mkt}}$

Thus the CML & the efficient frontier are tangential?